Choosing Chunk Size and Overlap for an Engineering Curriculum RAG Dataset

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***Abstract*—**I compared chunk size and overlap using the engineering PDF from my PT-F1 activity. I tested the 1,000-character, 200-character-overlap baseline and twelve configurations across 300, 500, 1,000 and 1,400 characters. I kept the dataset, embedding model, nine questions and Top-K of three unchanged. I measured Precision@3, evidence-unit Recall@3 and MRR@3 using fixed supporting facts. The run produced 13 indexes, 117 question tests and 351 retrieved records. C10, with 1,400 characters and no overlap, had the highest precision (0.2593) and MRR (0.4815). C11, with 140-character overlap, had slightly higher recall (0.4815 versus 0.4722) and fully covered four questions instead of three. But C10 performed much better on difficult questions and used only 50 chunks. I recommend C10 as a provisional setting for this dataset. The low overall scores, small test set and embedding truncation limit how far I can generalize this result.

***Index Terms*—**retrieval-augmented generation, chunking, FAISS, evidence recall, curriculum information

I. Introduction

Curriculum questions need careful source matching. CPE C405 and ECE C405 refer to different courses, so familiar words alone do not establish relevance. I tested chunk boundaries to see which settings preserved the facts needed to answer these questions.

Retrieval-augmented generation combines document retrieval with language generation [1]. For this activity, I focused on the retrieval stage. I wanted to find which chunk size and overlap preserved useful evidence without producing unnecessary chunks. I followed the PT-F2 activity requirements [2] and extended my PT-F1 notebook rather than changing its embedding model, language model or prompt.

I obtained these results from my executed PTF2_BERNARDO.ipynb notebook [3]. I transferred its exported tables into Excel. The PDF supplies the supporting facts; the notebook supplies the experiment measurements.

II. Dataset and Test Questions

A. Document and source characteristics

I used the same 26-page Computer Engineering and Electronics Engineering excerpt from PT-F1 [4]. It contains Computer Engineering and Electronics Engineering curricula, group-requirements tables, course descriptions and program policies. Similar course codes, repeated unit values and facts spread across sections make source identity important.

I extracted text by PDF page and replaced repeated whitespace with single spaces. I kept the one-based PDF-page position and each chunk's start and end offsets in the normalized text. I also recorded printed page numbers in the evidence workbook. For example, PDF page 25 is printed page 198. These are two references to the same page, not interchangeable page counts.

The PT-F1 materials identify the source as SY 2026–2027, and the retained chatbot prompt uses that description. But the filtered file has no handbook cover, so I cannot independently confirm its edition from this excerpt. I treat my findings as results for this supplied file.

B. Question coverage and required facts

I retained Q1–Q5 from PT-F1 and added Q6–Q9 before the latest configuration run. The set has three Easy, three Average and three Difficult questions. Easy questions ask for CPE C401 credit units, CPE C402 internship hours and the CPE C404 prerequisite. Average questions ask about CPE C402 prerequisites and hours, the BSEcE probation policy, and ECE C401 prerequisites and units.

The difficult questions compare ECE C405 with ECE C406, distinguish CPE C405 from ECE C405, and compare three conflicting unit listings for Computer Architecture and Organization. I used these questions to test whether retrieval could preserve course identity and collect evidence from different sections.

The nine questions require 1, 1, 2, 4, 4, 1, 4, 6 and 3 evidence units, respectively. This gives 26 units in total. An evidence unit is one required fact, such as a course's complete prerequisite set. Several units can be supported by one chunk. The notebook and workbook retain the exact questions, answers and source clauses.

III. Experimental Methodology

A. Configurations and controls

I compared the required baseline of 1,000 characters with 200-character overlap against C1–C12. The alternatives combine chunk sizes of 300, 500, 1,000 and 1,400 characters with overlap rates of 0%, 10% and 20%. Table I gives every tested value. C9 repeats the baseline settings, but its chunks and index were rebuilt separately.

I split each page independently. The start position advances by chunk size minus overlap, so a chunk cannot cross a PDF-page boundary. I checked that overlap was smaller than chunk size and that every saved chunk matched the normalized page slice at its recorded offsets. I built a new index for every configuration and used the same nine questions in the same order with $k = 3$.

I retained `all-MiniLM-L6-v2` for both document and query embeddings. It produces 384-dimensional vectors [5]. I normalized the vectors, converted them to float32 and used a FAISS `IndexFlatIP` index. Inner products of these unit-length vectors give cosine-similarity rankings. FAISS provides the similarity-search implementation [6]. I did not add a reranker or change retrieval settings between configurations.

I ran the notebook in Google Colab with a Tesla T4 GPU. The embedding device was `cuda:0`; the FAISS index used `faiss-cpu`. I retained Qwen3-8B in non-thinking mode with 4-bit NF4 double quantization, float16 computation, seed 42, a 4,096-token input budget and a 256-token output budget. The original source-grounding prompt was unchanged. These language-model settings preserve the PT-F1 pipeline, but generated answers do not enter the retrieval scores.

B. Relevance and evidence matching

I marked a chunk relevant when it contained at least one complete required fact and linked that fact to the correct course or section. A topic match alone did not count. For course-description facts, the primary rule required the course heading and the complete supporting clause within the retrieved chunk. A prerequisite set counted as one unit only when all its required codes were present.

Some facts also appear in curriculum tables or in equivalent wording within a course description. After reviewing an earlier run, I added four explicit rules for these occurrences. I then reran all configurations with those rules unchanged. The extra rules accept ECE C401's credit units and selected CPE/ECE C405 titles and credit units. They do not fill in prerequisites or other facts missing from a retrieved passage. The workbook's Evidence tab identifies all four passages and the facts each can support.

This rule-development history matters. I fixed the evidence units and matching rules before the latest run, but the rules were informed by earlier results on the same questions. I therefore treat this as a development experiment, not an evaluation on an untouched test set.

C. Metrics and automatic checks

For question q, I let $r_{q,i}$ be 1 when retrieved rank i is relevant and 0 otherwise. I let E_q be its fixed required-unit set and $S_{q,i}$ the units supported by that chunk. I calculated

$$P_q@3 = \frac{\sum_{i=1}^{3} r_{q,i}}{3}, \qquad R_q@3 = \frac{\left|\bigcup_{i=1}^{3} S_{q,i}\right|}{|E_q|}. \tag{1}$$

Recall counts each required fact once, even if two retrieved chunks repeat it. I calculated reciprocal rank as

$$RR_q@3 = \begin{cases} 1/\min\{i : r_{q,i} = 1\}, & \text{if a relevant rank exists,} \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

I averaged the nine question scores for overall precision, recall and MRR. I also averaged the three questions within each difficulty. A fully covered question has recall equal to one. This is evidence-unit recall, not recall over every potentially relevant chunk in the document.

The notebook checked all 351 retrieved records. It verified row counts, unique IDs, ranks 1–3, source-text offsets, valid evidence IDs and agreement between stored labels and the declared matcher. All records passed those checks. This confirms consistency with the scoring procedure; it does not replace independent human relevance review or mean every retrieved chunk is relevant.

D. Efficiency measurements

I recorded chunk count, indexing time, query latency and serialized FAISS index size. Indexing time includes chunking, embedding and index construction after model loading. Retrieval time includes query embedding and index search. I synchronized CUDA at timing boundaries when needed and calculated mean query latency across the nine queries, counting each query once rather than once per retrieved row.

Token-length diagnostics and file exports were outside indexing timing. I measured each configuration once, so the timing values describe this run. They do not establish a stable speed advantage across sessions.

IV. Results

A. Overall retrieval quality

C10 had the highest Precision@3 at 0.2593 and the highest MRR@3 at 0.4815. Its Recall@3 was 0.4722. C11 had the highest recall at 0.4815, but its precision fell to 0.1852 and its MRR to 0.4074. The recall gap was only 0.0093, while C10's precision and MRR were each 0.0741 higher. So recall alone did not settle my choice.

Compared with the baseline, C10 improved precision from 0.2222 to 0.2593, recall from 0.3981 to 0.4722, and MRR from 0.3704 to 0.4815. It used 50 chunks instead of 84, about 40.5% fewer. C9 reproduced the baseline's quality scores exactly, which supports consistency for that repeated setting. Its measured times were different.

C10 retrieved at least one relevant chunk for six of nine questions and fully covered three. C11 retrieved relevant evidence for five questions but fully covered four. C10 covered 15 of the 26 individual required units across the question set, while C11 covered 10. These pooled counts help explain their different strengths, but I did not substitute them for macro-average recall. The reported recall gives each question equal weight.

TABLE I
CONFIGURATION-LEVEL RESULTS FROM THE EXECUTED NOTEBOOK. SCORES ARE MACRO-AVERAGES OVER NINE QUESTIONS. FULL MEANS ALL REQUIRED EVIDENCE WAS RETRIEVED.

Config.	Size (characters)	Overlap (characters)	Overlap (%)	Chunks	Precision@3	Recall@3	MRR@3	Full (/9)
Base	1000	200	20	84	0.2222	0.3981	0.3704	3
C1	300	0	0	201	0.1481	0.1204	0.1296	0
C2	300	30	10	220	0.0741	0.0833	0.0556	0
C3	300	60	20	246	0.0370	0.0278	0.0556	0
C4	500	0	0	126	0.1111	0.1759	0.1852	1
C5	500	50	10	135	0.1852	0.2130	0.2778	1
C6	500	100	20	153	0.1852	0.1852	0.2593	0
C7	1000	0	0	69	0.1111	0.1759	0.1667	1
C8	1000	100	10	74	0.1852	0.2130	0.2593	1
C9	1000	200	20	84	0.2222	0.3981	0.3704	3
C10	1400	0	0	50	0.2593	0.4722	0.4815	3
C11	1400	140	10	62	0.1852	0.4815	0.4074	4
C12	1400	280	20	66	0.1852	0.3333	0.3333	2

TABLE II
PERFORMANCE BY DIFFICULTY. EACH GROUP CONTAINS THREE QUESTIONS. BASE IS THE BASELINE; P, R AND M MEAN PRECISION@3, RECALL@3 AND MRR@3.

	Easy			Average			Difficult		
Config.	P	R	M	P	R	M	P	R	M
Base	0.1111	0.3333	0.3333	0.4444	0.7500	0.6111	0.1111	0.1111	0.1667
C1	0.0000	0.0000	0.0000	0.3333	0.2500	0.2778	0.1111	0.1111	0.1111
C2	0.0000	0.0000	0.0000	0.2222	0.2500	0.1667	0.0000	0.0000	0.0000
C3	0.0000	0.0000	0.0000	0.1111	0.0833	0.1667	0.0000	0.0000	0.0000
C4	0.0000	0.0000	0.0000	0.2222	0.4167	0.4444	0.1111	0.1111	0.1111
C5	0.0000	0.0000	0.0000	0.3333	0.4167	0.5000	0.2222	0.2222	0.3333
C6	0.0000	0.0000	0.0000	0.3333	0.3333	0.4444	0.2222	0.2222	0.3333
C7	0.0000	0.0000	0.0000	0.2222	0.4167	0.3333	0.1111	0.1111	0.1667
C8	0.0000	0.0000	0.0000	0.3333	0.4167	0.4444	0.2222	0.2222	0.3333
C9	0.1111	0.3333	0.3333	0.4444	0.7500	0.6111	0.1111	0.1111	0.1667
C10	0.1111	0.3333	0.3333	0.2222	0.4167	0.5000	0.4444	0.6667	0.6111
C11	0.2222	0.6667	0.6667	0.2222	0.6667	0.4444	0.1111	0.1111	0.1111
C12	0.1111	0.3333	0.3333	0.1111	0.3333	0.3333	0.3333	0.3333	0.3333

B. Performance by difficulty

C10 was the strongest setting for difficult questions. Its difficult-question precision, recall and MRR were 0.4444, 0.6667 and 0.6111. C11 scored 0.1111 on all three. C12 had the second-highest difficult-question recall at 0.3333 and tied several settings for the next-highest MRR at 0.3333. Both were well below C10. This comparison matters because the difficult questions require course disambiguation and several facts.

C11 was stronger on Easy questions, with recall and MRR of 0.6667. C10 scored 0.3333 on both. The baseline and C9 were strongest on Average questions, with recall 0.7500 and MRR 0.6111. No setting led every difficulty group, so I do not describe C10 as uniformly best across all questions.

Q1 had no relevant top-three result under any configuration. Even the recommended setting therefore failed one direct factual question throughout this test. Table II and Fig. 1 show improvements within a retrieval system that still needs work, not complete coverage.

C. Effect of size at the same overlap percentage

At 0% overlap, 1,400 characters performed best on all three overall metrics. At 10%, 1,400 characters again had the highest recall and MRR. Its precision of 0.1852 tied the 500- and 1,000-character settings. At 20%, 1,000 characters performed best on all three metrics: precision 0.2222, recall 0.3981 and MRR 0.3704.

I took this to mean that a larger chunk was useful, but its benefit depended on the boundary pattern created by overlap. The results do not support a rule that increasing size always improves retrieval.

D. Effect of increasing overlap

At 300 characters, increasing overlap from 0% to 20% raised chunk count from 201 to 246, while recall fell from 0.1204 to 0.0278. At 500 characters, 10% overlap improved recall from 0.1759 to 0.2130, but 20% reduced it to 0.1852. More repeated text did not consistently preserve more useful evidence.

The 1,000-character group benefited from overlap in this run: recall rose from 0.1759 at 0% to 0.3981 at 20%, while chunk count rose from 69 to 84. For 1,400 characters, 10% overlap gave a small recall gain over zero overlap, but precision and MRR declined. At 20%, recall fell to 0.3333. Overlap increased the chunk count for every tested size, but the quality effect varied.

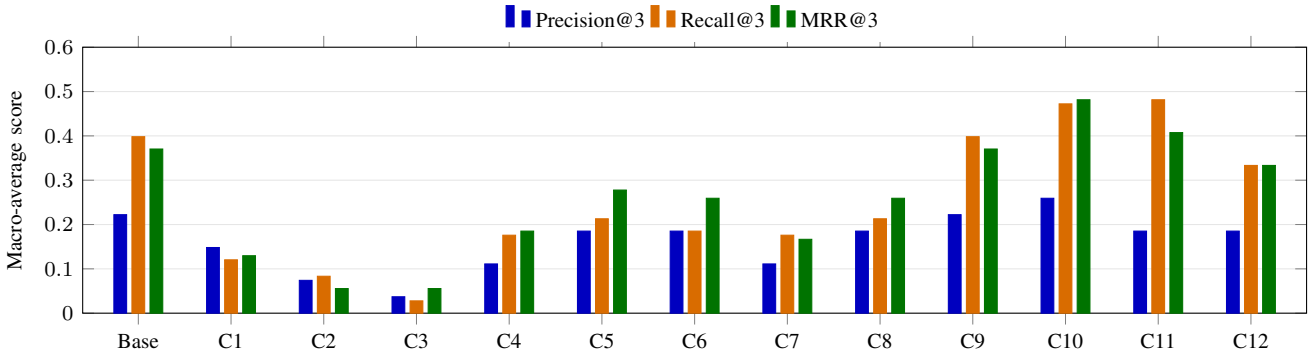


Fig. 1. The three retrieval metrics for the baseline and C1–C12, calculated from the notebook results. Higher values are better.

TABLE III
MEASURED EFFICIENCY. QUERY TIME IS THE MEAN ACROSS NINE QUESTIONS. OVER-LIMIT COUNTS CHUNKS EXCEEDING THE EMBEDDING MODEL'S 256-TOKEN LIMIT.

Config.	Indexing (s)	Query (ms)	Index (KiB)	Over-limit
Base	0.397	22.278	126.044	3
C1	0.188	6.693	301.544	0
C2	0.198	6.707	330.044	0
C3	0.218	7.175	369.044	0
C4	0.176	6.839	189.044	0
C5	0.180	6.785	202.544	0
C6	0.214	6.680	229.544	0
C7	0.177	6.853	103.544	3
C8	0.190	7.093	111.044	3
C9	0.205	8.307	126.044	3
C10	0.141	7.006	75.044	32
C11	0.175	6.651	93.044	36
C12	0.187	6.464	99.044	38

V. DISCUSSION

A. When larger context helped

The clearest success was C10 on Q5, which asks how ECE C405 and ECE C406 differ in prerequisites and lecture/laboratory units. Retrieved rank 2 was chunk 47 on PDF page 25, printed page 198. It contained both course headings and all four required facts: C405's three prerequisite codes and one lecture unit, plus C406's C405 prerequisite and one laboratory unit.

Only that rank was relevant for Q5. The resulting precision was $1/3$, recall was $4/4 = 1$, and reciprocal rank was $1/2$. This example showed that a lower precision can still accompany complete evidence when one chunk contains several required facts. The same chunk also included neighboring course material, so larger context was useful without being entirely focused.

C10 also covered four of Q8's six units across ranks 1 and 2. It retrieved titles and credit units for the two C405 courses, but not their complete prerequisite evidence. Precision was $2/3$, recall was $4/6$, and reciprocal rank was 1. This was a useful partial retrieval, not a fully answered comparison.

B. When small chunks lost identity

C1's top result for Q3 was chunk 133 on PDF page 18, at normalized offsets 3000–3060. It contained the three prerequisite codes and the 240-hour phrase, but started inside the word "Prerequisite" and omitted the CPE C402 heading. Its similarity score was 0.8499, yet it did not pass the source-identity rule.

I found this example useful because it separates similarity from sufficient evidence. A short passage can rank highly while losing the identity needed to attach its facts to the right course. Under the fixed convention, C1 received zero on all three metrics for Q3. The strict rule may reject a fragment that a reader could connect using outside context, but that context was not in the retrieved chunk itself.

Small chunks did not improve Easy-question performance in this run. All 300- and 500-character settings scored zero for that group. I therefore found no support here for choosing small chunks just because the question asks for a focused fact.

C. Quality and efficiency trade-offs

C10 combined the best precision and MRR with the fewest chunks and smallest index, 75.044 KiB. Its indexing time was 0.141 s. C11 used 62 chunks, a 93.044 KiB index and 0.175 s for indexing. C11's mean query time was slightly lower, 6.651 ms versus C10's 7.006 ms. C12 had the lowest observed query time, 6.464 ms, but lower retrieval quality. I did not treat these small latency gaps as decisive.

C5 and C8 show a different trade-off. They had identical overall precision (0.1852) and recall (0.2130), but used 135 and 74 chunks. Their MRR values were close, 0.2778 and 0.2593. C8 therefore offered similar coverage with about 45.2% fewer chunks. This showed that storage cost can differ substantially even when two quality measures match.

The baseline and C9 had identical settings and quality, but indexing took 0.397 s and 0.205 s, respectively. Their mean query times were 22.278 ms and 8.307 ms. The baseline ran first, so startup effects may explain part of the gap. I did not measure warm-up separately. Chunk count and serialized index size are firmer efficiency comparisons than these single-run timings.

D. Limitations

The largest technical limitation is embedding truncation. The model's default limit is 256 word pieces [5], which the run also recorded as its maximum sequence length. C10 had 32 of 50 chunks above that limit. C11 had 36 of 62 and C12 had 38 of 66. The 1,000-character settings each had three over-limit chunks, while the smaller settings had none.

The notebook scored the full stored chunk text, but the embedding model could represent only the retained token prefix. So a fact could count as retrieved evidence even if it was beyond the text represented in the vector. My results describe this exact pipeline. They do not show what would happen if every large chunk were embedded in full. A token-aware chunker or a longer-context embedding model would need a separate controlled test.

The nine questions meet the activity minimum, but each difficulty group contains only three. One changed result can therefore move a group average considerably. I used one PDF, one embedding model and one observation per timing measurement. The model revision field in the saved manifest was unavailable, which also limits exact environment reconstruction.

My labels were produced by explicit matching rules, with automatic consistency checks. The rules require complete facts and course identity, and I refined four rules after an earlier run. An independent relevance review and new questions are needed to check whether the recommendation holds beyond this development set. The source also contains conflicting unit values, and extraction flattens tables and retains some awkward spacing. I preserved those issues rather than assuming corrections. Finally, I evaluated retrieval, not generated-answer correctness or groundedness.

VI. Final Recommendation

I recommend C10: 1,400-character chunks, zero overlap and Top-K 3 for this dataset and the retained pipeline. It improved all three overall quality metrics over the baseline, led precision and MRR, and performed best on difficult questions. It also used only 50 chunks, which reduced embedding work and index storage compared with the other settings.

I considered C11 carefully because its recall was slightly higher and it fully covered four questions rather than three. But its difficult-question recall dropped from C10's 0.6667 to 0.1111. C10 also found some supporting evidence for six questions rather than five. For a dataset with similar course names and comparisons across sections, I value that wider partial coverage and stronger difficult-question performance. This makes C10 my practical choice without claiming that it wins every metric or every question.

The recommendation is provisional because C10 fully covered only three questions. Before chatbot use, I would test new questions, review labels independently, control embedding truncation and repeat timings after warm-up. C11 remains useful when Easy- and Average-question completeness is the priority.

VII. Reproducibility and Submitted Materials

My documented PTF2_BERNARDO.ipynb notebook contains the dataset-upload step, original pipeline, fixed questions, evidence units, matching rules, configuration loop, metric calculations, chart and row checks. Its saved outputs show the results directly. The accompanying workbook contains the configuration results, difficulty results, detailed 351-row retrieval log, question calculations and source evidence. The source dataset is the previously supplied PT-F1 file, PTF1_Member1_Computer_and_Electronics_Engineering.pdf. Its verified evidence is recorded in the notebook and workbook, as required by the activity [2].

To reproduce the full experiment, I would run the notebook from the dependency cell, restart the Colab session if needed, upload the same PDF and execute through Component III, Section 3.7. The source file is checked against SHA-256

```
0303c2ee0e1b253bfabc33a73e8053cf08c342ed3643
566b92418cda1cc3b19f
```

The configuration loop rebuilds all 13 indexes. Section 3.6 verifies every retrieved row. Section 3.7 displays the final tables whether or not the optional download flag is enabled. Setting `EXPORT_RESULTS=True` saves the current tables and supporting files; setting it back to False only skips that download step.

After the full run, the following code identifies my final setting and prints its measured scores. It checks the notebook results rather than replacing the experiment loop.

```
FINAL_CONFIG = "C10"
assert EXPERIMENTS[FINAL_CONFIG] == (1400, 0, 3)
result = TABLE_A.set_index("Configuration")
print(result.loc[FINAL_CONFIG, [
    "Precision_at_3", "Recall_at_3", "MRR_at_3",
    "Total_Chunks", "Full_Questions"
]])
```

The saved software record lists PyTorch 2.11.0+cu130, Transformers 5.22.1, Sentence Transformers 6.2.0, faiss-cpu 1.15.1, pypdf 6.21.4, NumPy 2.1.3 and pandas 2.2.3. With the same source text, settings, software and rules, I expect the retrieval labels to be consistent. Timing values can change across Colab sessions, and I do not promise bit-for-bit identical measurements on different hardware.

References

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